PMGSY – Physical and Financial Performance Analysis.

Chemirthi Palli Thanush
Computer Science and Engineering
Lovely Proffessional University
line 4: Jalandhar, Indai
thanushthanu0610@gmail.com

***Abstract*—This project focuses on analyzing the Pradhan Mantri Gram Sadak Yojana (PMGSY) using Power BI to evaluate both physical and financial performance of rural road infrastructure projects across India. The study uses data collected from the official government portal** data.gov.in **and transforms raw datasets into interactive dashboards for meaningful analysis. Key parameters such as sanctioned road length, completed road length, expenditure incurred, cost per kilometer, and bridge construction were examined through various visualizations. The project highlights state-wise and scheme-wise performance differences, identifies high-performing and low-performing regions, and reveals that higher expenditure does not always guarantee better completion rates. Power BI features such as slicers, drill-downs, KPI cards, and maps were used to create an efficient decision-support system. The findings of this project can help policymakers and administrators improve planning, monitoring, resource allocation, and implementation of rural infrastructure schemes.**

Keywords—PMGSY, Power BI, Data Analytics, Rural Road Infrastructure, Dashboard Visualization, Financial Performance, Physical Progress, State-wise Analysis

I. Introduction (*Heading 1*)

PMGSY (Pradhan Mantri Gram Sadak Yojana) is a government scheme launched to improve rural road connectivity across India. Good road infrastructure helps villages access education, healthcare, markets, and employment opportunities. This project uses Power BI to analyze PMGSY data related to road construction and financial performance. The study provides interactive dashboards to support better planning, monitoring, and decision-making.

II. Ease of Use

A. User-Friendly Interfacea

The Power BI dashboard was developed with a simple and interactive interface to ensure ease of use for all users. It allows users to navigate through charts, tables, and KPI cards without requiring advanced technical knowledge. Filters and slicers are provided for quick access to state-wise, district-wise, and scheme-wise information. This makes data exploration smooth and efficient.

B. Easy Navigation and Accessibility

The dashboard maintains a clear layout with properly arranged visuals, labels, and controls for better accessibility. Users can easily switch between different views, apply filters, and analyze performance with minimal effort. Interactive features such as drill-down, cross-filtering, and highlighting improve usability. This design helps policymakers and planners make faster and better decisionsand not as an independent document. Please do not revise any of the current designations.

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III. Prepare Your Paper Before Styling

Before developing the Power BI dashboard, the dataset was first collected, cleaned, and organized in a separate workspace. All necessary corrections, transformations, and validation were completed before creating visuals and reports. Raw data files and processed data were managed separately to maintain accuracy and consistency. Proper planning and preparation ensured smooth dashboard development and reliable analysis results..

A. Data Cleaning and Preparation

The PMGSY dataset was examined for missing values, duplicate records, and inconsistent entries before analysis. State names, district names, and scheme categories were standardized for accurate grouping. Numerical fields such as road length and expenditure were converted into proper formats. This step improved data quality and reliability.

B. Standardization of Units

- All financial values were maintained in lakhs, while road length values were kept in kilometers for consistency. Standard units help in clear comparison across states and schemes. Mixing multiple units was avoided to prevent confusion in calculations and reporting. This ensured meaningful and accurate analysis.

C. Validation and Proofreading

The dataset was carefully reviewed to identify errors and unusual values before loading into Power BI. Aggregated values were cross-checked with source records for correctness. Proper validation reduced inaccuracies in dashboard KPIs and charts. This step strengthened the credibility of the project findings.

D. Structured Organization

- The project content was organized into sections such as physical progress, financial performance, scheme-wise analysis, and state-wise comparison. Maintaining a structured workflow made dashboard creation easier and more efficient. Clear organization also improved readability and presentation quality.
- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.

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IV. Using the Template

After completing data cleaning and analysis, the project dashboard was developed using Power BI templates and standard visualization practices. A structured layout was followed to present insights clearly and professionally. Prepared datasets were imported into Power BI, where charts, KPI cards, and filters were applied. This process ensured consistency, readability, and effective presentation of PMGSY analysis results.

A. Authors and Affiliations

a) The project report includes the student name, registration number, section, course code, and faculty guide details. Proper author and affiliation information helps identify ownership and academic supervision of the project. This also adds professionalism and authenticity to the report.

B. Identify the Headings

Headings are used to organize the report into logical sections such as Introduction, Objectives, Data Source, Analysis, Dashboard Design, and Conclusion. Clear headings improve readability and guide the reader through the project systematically. Subheadings were also used for detailed analysis topics like physical and financial performance.

C. Figures and Tables

Various figures and charts such as bar graphs, scatter plots, maps, and KPI cards were used in the dashboard to represent data visually. Tables were used where numerical comparisons were necessary. Proper placement of visuals after relevant explanations improves understanding and makes the report more attractive

The above dashboard provides a comprehensive overview of the physical progress of the **Pradhan Mantri Gram Sadak Yojana (PMGSY)** by presenting key performance indicators, state-wise comparisons, expenditure trends, and road completion status. The dashboard was designed in Power BI with an interactive dark theme layout to make the information visually attractive and easy to interpret. It helps policymakers and planners evaluate project implementation efficiency across India

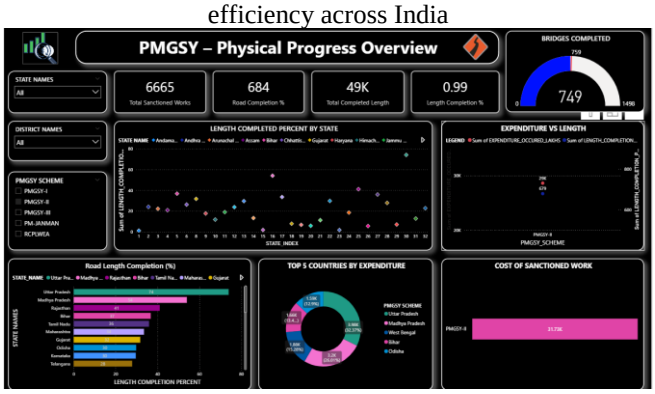

Fig. 1. Overall Dashboard

1) 1. Key Performance Indicators (KPIs)

The top section of the dashboard highlights major summary indicators related to PMGSY physical progress. These KPIs provide an instant understanding of the scale and performance of the scheme.

- **Total Sanctioned Works: 6665**
  This indicates the total number of road construction works approved under PMGSY across selected states and schemes. It reflects the overall infrastructure development scope under the program.
- **Road Completion %: 684**
  This value represents the aggregated road completion progress based on selected filters. It helps assess how effectively sanctioned works are being converted into completed roads.
- **Total Completed Length: 49K**
  This shows that approximately 49,000 kilometers of road length have been completed under PMGSY. It reflects the physical achievement of the project.
- **Length Completion %: 0.99**
  This suggests that road completion is close to the sanctioned target, indicating efficient implementation in many regions.
- **Bridges Completed: 749**
  The gauge chart displays the total number of bridges completed out of the sanctioned bridge target. Bridge development is crucial in regions with rivers, valleys, and difficult terrain.

2) State, District, and Scheme Filters

The left panel contains interactive slicers for:

- State Names
- District Names
- PMGSY Scheme Categories (PMGSY-I, PMGSY-II, PMGSY-III, PM-JANMAN, RCPLWEA)

These filters allow users to dynamically analyze specific states, districts, or schemes. This feature improves dashboard usability and supports customized decision-making.

3) Length Completed Percent by State (Scatter Plot)

This scatter plot compares completion percentage across states using state index values. Each point represents a state.

a) Analysis:

- Considerable variation exists among states.
- Some states show high completion percentages, indicating strong implementation capacity.
- Other states remain at lower levels, suggesting delays or operational challenges.

This visual is useful for identifying high-performing and low-performing states quickly.

4) 4. Expenditure vs Length Analysis

This dual-axis line chart compares:

- Expenditure incurred (Lakhs)
- Length completed

across PMGSY schemes.

a) Analysis:

- PMGSY-II shows the highest expenditure and substantial road completion.
- Higher expenditure does not always guarantee proportionately higher road completion.
- This indicates the need to monitor financial and physical performance together.

The chart helps evaluate cost efficiency of each scheme.

5) 5. Road Length Completion (%) – Bar Chart

This horizontal bar chart ranks top-performing states based on road completion percentage.

a) Top Performing States:

- Uttar Pradesh – 74%
- Madhya Pradesh – 54%
- Rajasthan – 41%
- Bihar – 37%
- Tamil Nadu – 36%

b) Interpretation:

States with better completion percentages likely demonstrate strong planning, timely fund utilization, and efficient execution systems.

6) 6. Top 5 States by Expenditure (Donut Chart)

This chart shows states receiving or utilizing the highest expenditure under PMGSY.

a) Top States Include:

- Uttar Pradesh
- Madhya Pradesh
- West Bengal
- Bihar
- Odisha

b) Interpretation:

These states likely have larger rural populations, extensive road requirements, and higher infrastructure demand.

7) Cost of Sanctioned Work

The final chart indicates the cost of sanctioned work under schemes. PMGSY-II shows approximately **31.73K**, indicating substantial financial allocation.

a) Interpretation:

Higher sanctioned cost may be associated with broader project scope, larger road lengths, or higher construction difficulty.

8) Overall Insights

The dashboard reveals several important findings:

- PMGSY has achieved significant physical progress with 49K completed road length.
- Bridge construction is progressing steadily with 749 completed bridges.
- Performance varies widely across states.
- Some states show strong execution efficiency, while others need attention.
- Financial expenditure and physical completion are not always directly proportional.
- PMGSY-II appears to be one of the most resource-intensive schemes.

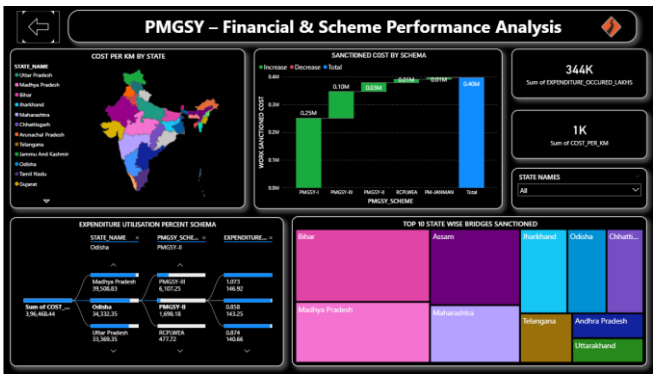

Fig. 2. This Figure tell about the Finacial State Wise

The above dashboard presents a detailed **Financial and Scheme-wise Performance Analysis** of the **Pradhan Mantri Gram Sadak Yojana (PMGSY)** using Power BI. It focuses on expenditure trends, sanctioned costs, cost efficiency, bridge approvals, and scheme contributions. The

dashboard is designed to help planners and policymakers evaluate how effectively funds are allocated and utilized across different states and PMGSY schemes.

D. 1. Dashboard Objective

The main purpose of this dashboard is to analyze the financial performance of PMGSY and compare the contribution of different schemes such as:

- PMGSY-I
- PMGSY-II
- PMGSY-III
- RCPLWEA
- PM-JANMAN

It provides an integrated view of expenditure, sanctioned cost, cost per kilometer, and infrastructure development indicators.

E. 2. Key Performance Indicators (KPIs)

Two KPI cards are displayed on the right side of the dashboard.

1) Total Expenditure Occurred: 344K

This represents the total amount spent on PMGSY road development projects. It indicates the scale of financial utilization across all selected states and schemes.

2) Sum of Cost per KM: 1K

This reflects the aggregated construction cost per kilometer of road development. It is useful for evaluating cost efficiency.

F. 3. Cost per KM by State (Map Visualization)

The filled map shows cost per kilometer across different states in India. Each color represents a state.

1) Analysis:

- States with difficult terrain or remote areas usually have higher road construction costs.
- Hilly and northeastern states may show increased costs due to transportation and engineering challenges.
- Plains states generally have lower construction costs due to easier accessibility and simpler terrain.

2) Importance:

This helps government agencies allocate funds realistically according to geographical conditions.

G. 4. Sanctioned Cost by Scheme (Waterfall Chart)

This chart compares sanctioned financial cost among PMGSY schemes.

1) Values Shown:

- PMGSY-I = 0.25M
- PMGSY-III = 0.10M
- PMGSY-II = 0.03M
- RCPLWEA = 0.01M
- PM-JANMAN = 0.01M
- Total = 0.40M

2) Interpretation:

- **PMGSY-I** has the highest sanctioned cost, showing strong focus on basic rural connectivity.
- **PMGSY-III** also receives substantial allocation, likely for upgradation and consolidation.
- **RCPLWEA** and **PM-JANMAN** receive smaller allocations as they are targeted schemes for specific regions.

H. 5. Expenditure Utilisation Percent Schema (Decomposition Tree)

This visual breaks down expenditure utilization across states and schemes.

1) Major States Highlighted:

- Madhya Pradesh
- Odisha
- Uttar Pradesh

2) Scheme Highlighted:

- PMGSY-II
- RCPLWEA

3) Analysis:

- Madhya Pradesh shows the highest expenditure contribution among selected states.
- Odisha and Uttar Pradesh also contribute significantly.
- PMGSY-II appears highly active in fund utilization.

This helps identify which states and schemes are using funds effectively.

I. 6. Top 10 State-wise Bridges Sanctioned (Treemap)

The treemap shows the top states with the highest number of bridges sanctioned.

1) Leading States Include:

- Bihar
- Madhya Pradesh
- Assam
- Maharashtra
- Jharkhand
- Odisha
- Telangana
- Andhra Pradesh
- Uttarakhand

2) Interpretation:

- Bihar has the highest bridge sanctions, indicating strong need for bridge infrastructure.
- States with rivers, flood-prone areas, and uneven terrain require more bridges for rural connectivity.

J. 7. State Filter

The right-side slicer allows users to select a particular state for customized analysis. Once selected, all charts update dynamically. This improves dashboard interactivity and user experience.

K. 8. Major Insights

- Total project expenditure is high at **344K**, showing significant government investment.
- PMGSY-I has the largest sanctioned cost share.
- Cost per kilometer varies by state due to terrain and accessibility.
- Madhya Pradesh, Odisha, and Uttar Pradesh are major expenditure-utilizing states.
- Bihar leads in bridge sanctions.
- Smaller schemes play a targeted but important role in difficult regions.

ACKNOWLEDGMENT

References include official sources such as data.gov.in, Ministry of Rural Development, and Microsoft Power BI documentation. Using trusted references increases the accuracy, reliability, and credibility of the project work.

REFERENCES

The references in this project are the official and trusted sources used to collect data, understand the PMGSY scheme, and build the Power BI dashboard. References are important because they increase the accuracy, reliability, and credibility of the report. They show that the analysis is based on authentic information rather than assumptions.

1) Government of India Open Data Portal (data.gov.in)

This is the primary source of the dataset used in the project. It provides official government data related to PMGSY road development, expenditure, sanctioned works, completed road length, and bridge construction. Since it is an authorized portal, the data is accurate and regularly updated.

Purpose in Project:

- Used for collecting PMGSY raw dataset
- State-wise and scheme-wise analysis
- Financial and physical performance evaluation